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wall. Normally, this is smaller than the impulsive effect, but if there is not enough dead load, the tank will tend to uplift.

9.2—Rectangular tanks (Type 1)

R9.2—Rectangular tanks (Type 1)

All equations in Section 9.2 except Eq. (9.2.4a), (9.2.4b), and (9.2.4c) were originally developed by Housner (1963a) and subsequently used by other authors (Housner 1956; NZS 3106; Haroun 1984; ASCE 1981; Veletsos and Shivakumar 1997; ANSI/AWWA D110-13 (R18); ANSI/AWWA D115-17; FEMA 450-2; ASCE/SEI 7-16; Haroun and Ellaithy 1985).

Equations (9.2.4a), (9.2.4b), and (9.2.4c) were adapted from NZS 3106.

9.2.1 Effective weights of accelerating liquid (Fig. 9.2.1)

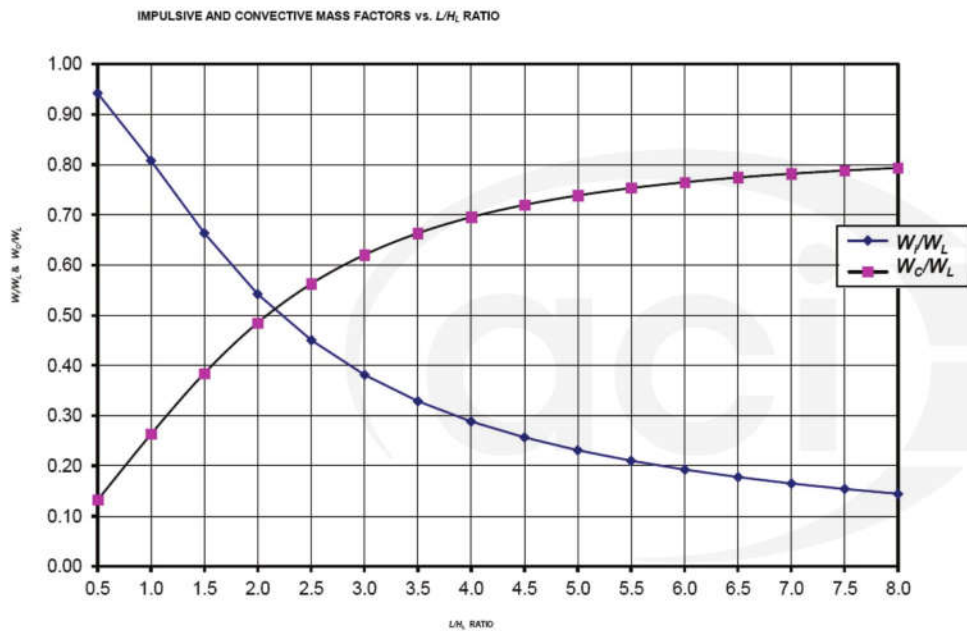


Fig. 9.2.1—Factors W_i/W_L and W_c/W_L versus ratio L/H_L for rectangular tanks.

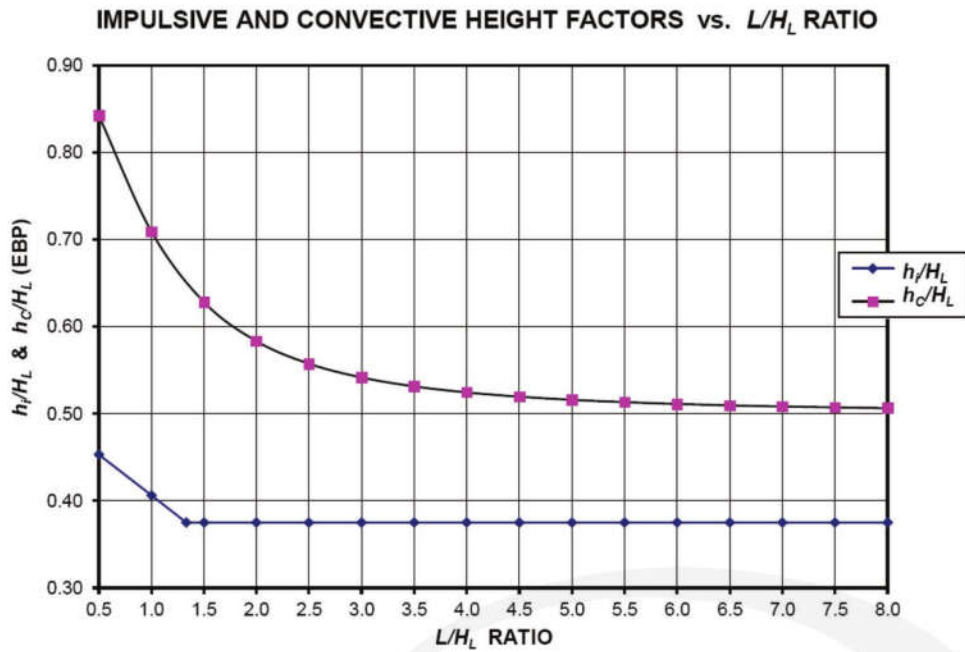
$$W_i/W_L = \frac{\tanh[0.866(L/H_L)]}{0.866(L/H_L)} \quad (9.2.1a)$$

$$W_c/W_L = 0.264(L/H_L)\tanh[3.16(H_L/L)] \quad (9.2.1b)$$

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9.2.2 Height to centers of gravity EBP (Fig. 9.2.2)

Fig. 9.2.2—Factors h_i/H_L and h_c/H_L versus ratio L/H_L for rectangular tanks (EBP).For tanks with $L/H_L < 1.333$

$$h_i/H_L = 0.5 - 0.09375(L/H_L) \quad (9.2.2a)$$

For tanks with $L/H_L \geq 1.333$

$$h_i/H_L = 0.375 \quad (9.2.2b)$$

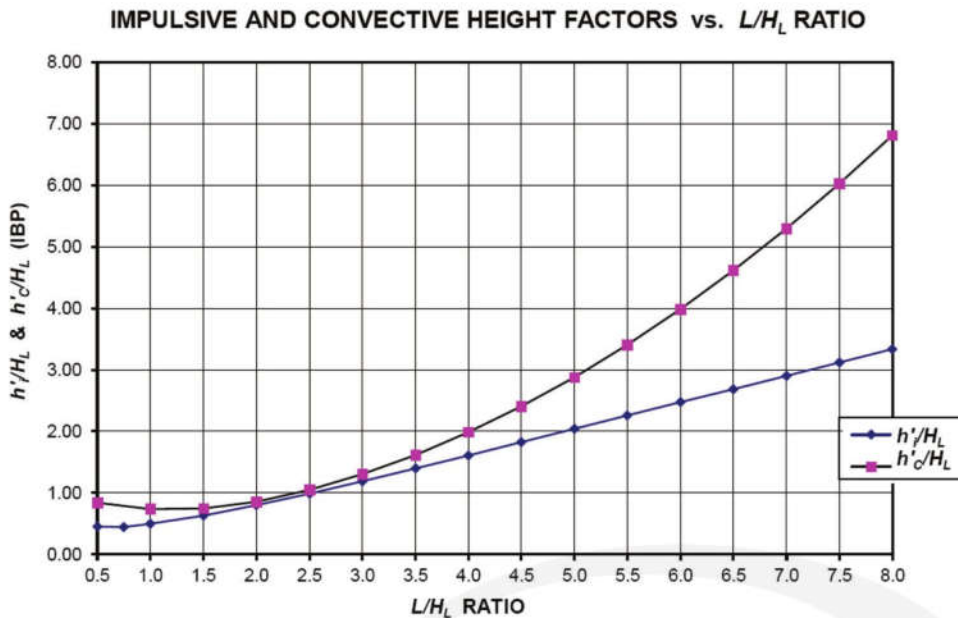
For all tanks

$$h_c/H_L = 1 - \frac{\cosh[3.16(H_L/L)] - 1}{3.16(H_L/L) \sinh[3.16(H_L/L)]} \quad (9.2.2c)$$

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9.2.3 Heights to center of gravity (IBP) (Fig. 9.2.3)

Fig. 9.2.3—Factors h_i'/H_L and h_c'/H_L versus ratio L/H_L for rectangular tanks (IBP).

For tanks with $L/H_L < 0.75$

$$h_i'/H_L = 0.45 \quad (9.2.3a)$$

For tanks with $L/H_L \geq 0.75$

$$h_i'/H_L = \frac{0.866(L/H_L)}{2 \tanh[0.866(L/H_L)]} - 0.125 \quad (9.2.3b)$$

For all tanks

$$h_c'/H_L = 1 - \frac{\cosh[3.16(H_L/L)] - 2.01}{3.16(H_L/L) \sinh[3.16(H_L/L)]} \quad (9.2.3c)$$

9.2.4 Dynamic properties—The structural stiffness k shall be computed on the basis of correct boundary conditions.

$$\omega_i = \sqrt{\frac{k}{m}} \quad (9.2.4a)$$

$$m = m_w + m_i \quad (9.2.4b)$$

$$T_i = \frac{2\pi}{\omega_i} = 2\pi \sqrt{\frac{m}{k}} \quad (9.2.4c)$$

$$\omega_c = \frac{\lambda}{\sqrt{L}} \quad (9.2.4d)$$

where

$$\lambda = \sqrt{3.16g \tanh[3.16(H_L/L)]} \quad (9.2.4e)$$

R9.2.4 Dynamic properties—The following equations are provided as examples for the special case of a wall of uniform thickness. (Note that mass is equal to weight divided by the acceleration due to gravity.)

$$m_w = H_w t_w \left(\frac{\gamma_c}{12g} \right) \quad (\text{in.-lb})$$

$$m_w = H_w t_w \left(\frac{\gamma_c}{g} \right) \quad (\text{SI})$$

$$m_i = \left(\frac{W_i}{W_L} \right) \left(\frac{L}{2} \right) H_L \left(\frac{\gamma_L}{g} \right) \quad (\text{in.-lb})$$

$$m_i = \left(\frac{W_i}{W_L} \right) \left(\frac{L}{2} \right) H_L \left(\frac{\gamma_L}{g} \right) \times 10^3 \quad (\text{SI})$$

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$$T_c = \frac{2\pi}{\omega_c} = \left(\frac{2\pi}{\lambda}\right) \sqrt{L} \quad (9.2.4f)$$

$\left(\frac{2\pi}{\lambda}\right)$ is from Fig. 9.2.4.

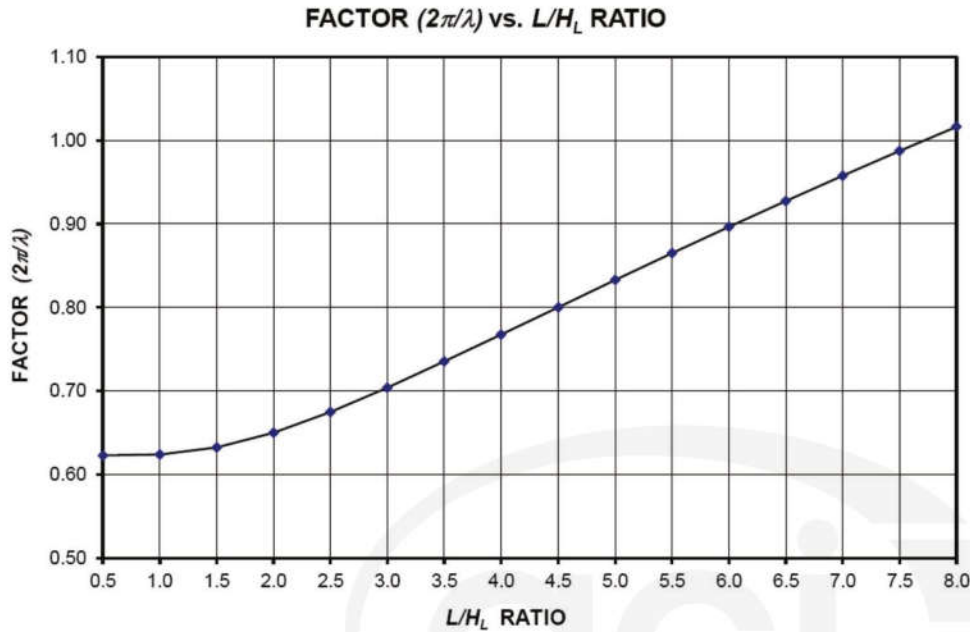


Fig. 9.2.4—Factor $2\pi/\lambda$ for rectangular tanks.

(For results in the SI system, multiply the factors on the vertical axis by 1.811.)

$$h = \frac{(h_w m_w + h_i m_i)}{(m_w + m_i)}$$

where $h_w = 0.5H_w$, and h_i is obtained from Eq. (9.2.2a) and (9.2.2b), and F_i .

For walls of nonuniform thickness, special analysis is required to determine m_w , m_i , and h .

For fixed-base, free-top cantilever walls such as in open-top tanks, flexural stiffness for a unit width of wall may be approximated using the following equation

$$k = \frac{E_c}{48} \left(\frac{t_w}{h} \right)^3 \quad (\text{in.-lb})$$

$$k = \frac{E_c}{4 \times 10^3} \left(\frac{t_w}{h} \right)^3 \quad (\text{SI})$$

Flexural stiffness formulas may be developed for other wall support conditions. Such spring constants will generally fall within the low period range (less than approximately 0.3 seconds) for tanks of normal proportions.

As an alternative to computing the natural period of vibration, particularly for end conditions other than cantilever, it is reasonable to assume the wall rigid. In such cases, Eq. (9.4.1b) may be conservatively used to calculate the impulsive forces regardless of the actual boundary conditions of the structure

or structural components being analyzed.

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9.3—Circular tanks (Type 2)

9.3.1 Effective weights of accelerating liquid (Fig. 9.3.1)

$$W_i/W_L = \frac{\tanh[0.866(D/H_L)]}{0.866(D/H_L)} \quad (9.3.1a)$$

$$W_c/W_L = 0.230(D/H_L)\tanh[3.68(H_L/D)] \quad (9.3.1b)$$

R9.3—Circular tanks (Type 2)

All equations in Section 9.3 except Eq. (9.3.4a) through (9.3.4f) were originally developed by Housner (1963a) and subsequently used by other authors (Housner 1956; NZS 3106; Haroun 1984; ASCE 1981; Veletsos and Shivakumar 1997; ANSI/AWWA D110-13 (R18); ANSI/AWWA D115-17; Haroun and Ellaithy 1985).

Equations (9.3.4a) through (9.3.4f) were adapted from NZS 3106.

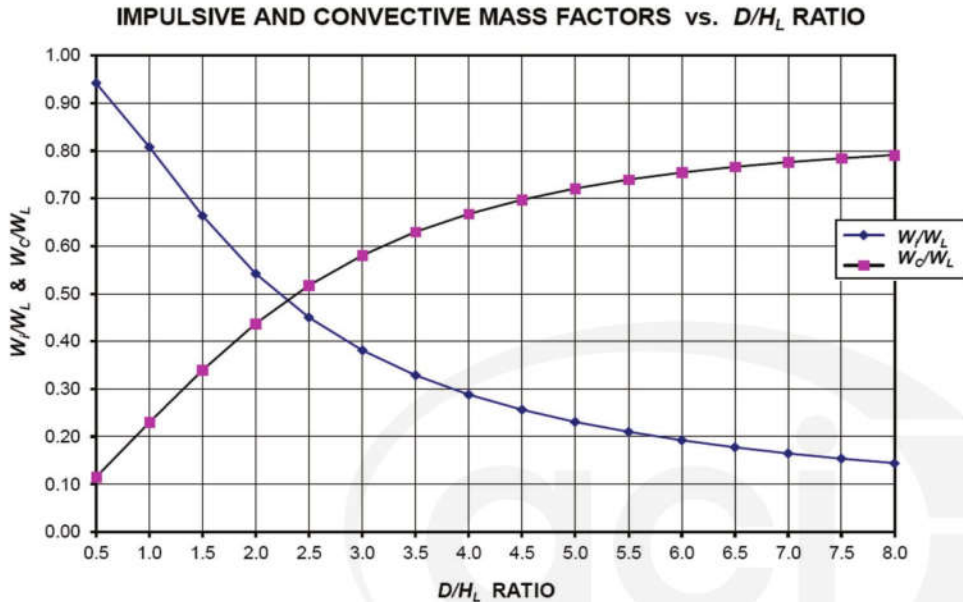


Fig. 9.3.1—Factors W_i/W_L and W_c/W_L versus ratio D/H_L for circular tanks.

9.3.2 Heights to centers of gravity (EBP) (Fig. 9.3.2)

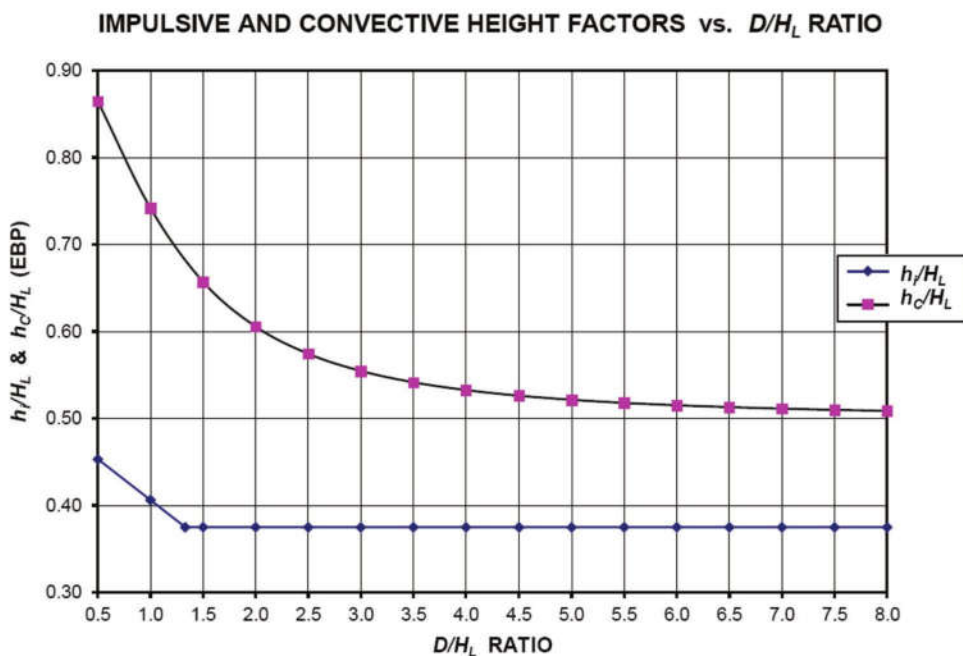


Fig. 9.3.2—Factors h_i/H_L and h_c/H_L versus ratio D/H_L for circular tanks (EBP).

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For tanks with $D/H_L < 1.333$

$$h_i/H_L = 0.5 - 0.09375(D/H_L) \quad (9.3.2a)$$

For tanks with $D/H_L \geq 1.333$

$$H_i/H_L = 0.375 \quad (9.3.2b)$$

For all tanks

$$h_c/H_L = 1 - \frac{\cosh[3.68(H_L/D)] - 1}{3.68(H_L/D) \sinh[3.68(H_L/D)]} \quad (9.3.2c)$$

9.3.3 Heights to center of gravity (IBP) (Fig. 9.3.3)

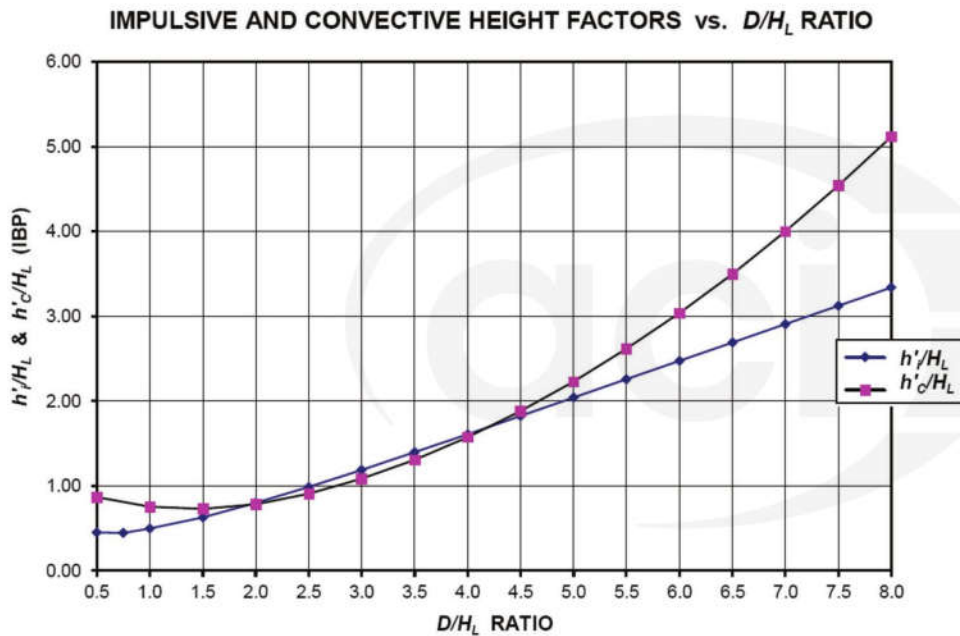


Fig. 9.3.3—Factors h_i'/H_L and h_c'/H_L versus ratio D/H_L for circular tanks (IBP).

For tanks with $D/H_L < 0.75$

$$h_i'/H_L = 0.45 \quad (9.3.3a)$$

For tanks with $D/H_L \geq 0.75$

$$h_i'/H_L = \frac{0.866(D/H_L)}{2 \tanh[0.866(D/H_L)]} - 0.125 \quad (9.3.3b)$$

$$h_c'/H_L = 1 - \frac{\cosh[3.68(H_L/D)] - 2.01}{3.68(H_L/D) \sinh[3.68(H_L/D)]} \quad (9.3.3c)$$

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9.3.4 Dynamic properties

For tank Types 2.1 and 2.2:

$$\omega_i = C_i \times \frac{12}{H_L} \sqrt{\frac{E_c g}{\gamma_c}} \quad (\text{in.-lb}) \quad (9.3.4a)$$

$$\omega_i = C_i \times \frac{1}{H_L} \sqrt{\frac{10^3 \times E_c g}{\gamma_c}} \quad (\text{SI})$$

$$C_i = C_w \times 10 \sqrt{\frac{t_w}{12r}} \quad (\text{in.-lb})$$

$$C_i = C_w \times \sqrt{\frac{t_w}{10r}} \quad (\text{SI}) \quad (9.3.4b)$$

C_w from Fig. 9.3.4a

COEFFICIENT C_w vs. D/H_L RATIO

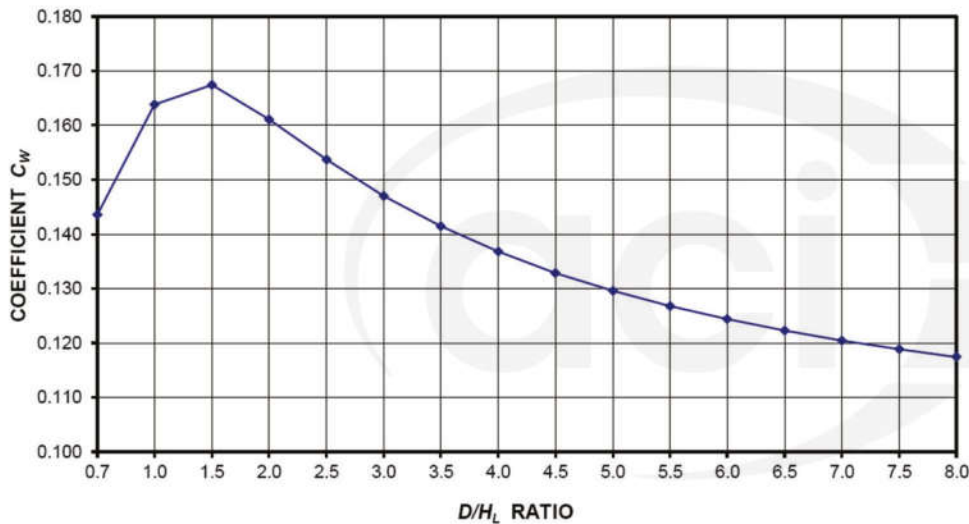


Fig. 9.3.4a—Coefficient C_w for circular tanks.

For $D/H_L > 0.667$

$$C_w = 9.375 \times 10^{-2} + 0.2039(H_L/D) - 1.034(H_L/D)^2 - 0.1253(H_L/D)^3 + 0.1267(H_L/D)^4 - 3.186 \times 10^{-2}(H_L/D)^5$$

$$T_i = 2\pi/\omega_i \quad (9.3.4c)$$

For tank Type 2.3

$$T_i = \sqrt{\frac{8\pi(W_w + W_r + W_i)}{gDk_a}} \quad (9.3.4d)$$

but shall not exceed 1.25s

R9.3.4 Dynamic properties—Equations (9.3.4a) and (9.3.4b) are adapted from **ASCE (1981)** and **Veletsos and Shivakumar (1997)**.

Equations (9.3.4d) and (9.3.4e) are adapted from **ANSI/AWWA D110-13 (R 18)** and **ANSI/AWWA D115-17**.

Equations (9.2.4e), (9.2.4f), (9.3.4g), and (9.3.4h) are adapted from **Housner (1963a)**.

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$$k_a = 144 \left[\left(\frac{A_s E_s \cos^2 \alpha}{L_c S_c} \right) + \left(\frac{2 G_p w_p L_p}{t_p S_p} \right) \right] \quad (\text{in-lb}) \quad (9.3.4e)$$

$$k_a = 10^6 \left[\left(\frac{A_s E_s \cos^2 \alpha}{L_c S_c} \right) + \left(\frac{2 G_p w_p L_p}{t_p S_p} \right) \right] \quad (\text{SI})$$

 T_c

$$\omega_c = \frac{\lambda}{\sqrt{D}} \quad (9.3.4f)$$

where

$$\lambda = \sqrt{3.68g \tanh[3.68(H_L/D)]} \quad (9.3.4g)$$

$$T_c = \frac{2\pi}{\omega_c} = \left(\frac{2\pi}{\lambda} \right) \sqrt{D} \quad (9.3.4h)$$

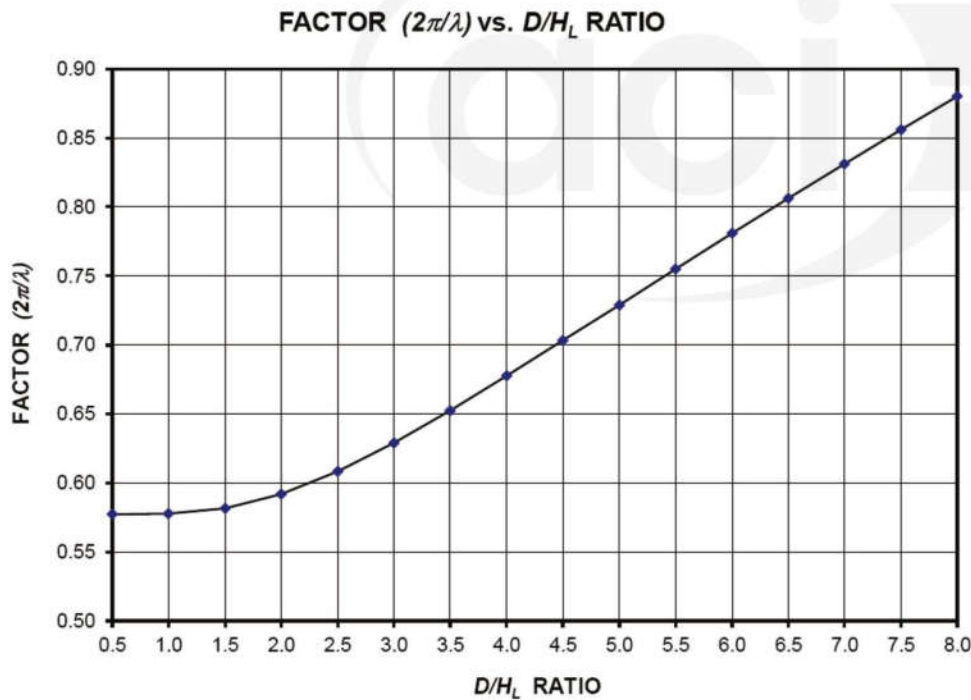
 $\left(\frac{2\pi}{\lambda} \right)$ from Fig. 9.3.4b


Fig. 9.3.4b—Factor $2\pi/\lambda$ for circular tanks.

(For results in the SI system, multiply the factors on the vertical axis by 1.811.)

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9.4—Seismic response coefficients C_i , C_c , and C_t R9.4—Seismic response coefficients C_i , C_c , and C_t

Designations C_i , C_c , and C_t define the profile of the design response spectrum at periods T_i , T_c , and T_v respectively. A plot of the seismic response coefficient C_i is shown in the design response spectrum in Fig. R9.4.1, which is adapted from **IBC-2003**.

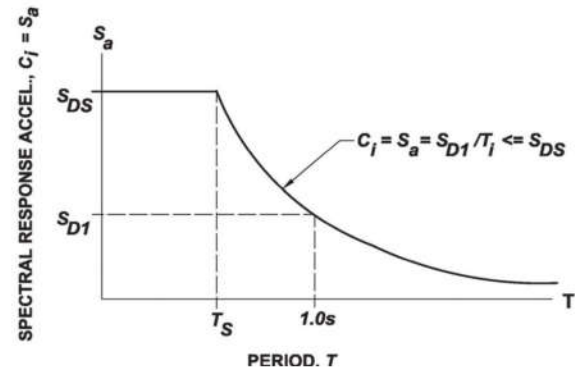


Fig. R9.4.1—Design response spectrum (5 percent damping).

9.4.1 C_i shall be determined as follows:

For $T_i \leq T_S$

$$C_i = S_{DS} \quad (9.4.1a)$$

For $T_i > T_S$

$$C_i = \frac{S_{D1}}{T_i} \leq S_{DS} \quad (9.4.1b)$$

where

$$T_S = \frac{S_{D1}}{S_{DS}} \quad (9.4.1c)$$

S_{DS} is the design spectral response acceleration at short periods.

$$S_{DS} = \frac{2}{3} S_s F_a \quad (9.4.1d)$$

S_{D1} is the design spectral response acceleration at 1 s period.

$$S_{D1} = \frac{2}{3} S_1 F_v \quad (9.4.1e)$$

S_s and S_1 are the mapped spectral response accelerations at short periods (S_s) and 1-second (S_1) respectively, and shall be obtained from the seismic ground motion maps 22-1 through 22-7 of **ASCE/SEI 7-16** Section 22; and F_a and F_v are the site coefficients and shall be obtained from Tables 11.4-1 and 11.4-2, respectively, of ASCE/SEI 7-16, in conjunction with Table 20.3-1, “Site Classification,” of ASCE/SEI 7-16.

R9.4.1 The mapped spectral response accelerations S_s and S_1 for any location can also be obtained from the latest database of the United States Geological Survey (USGS), at its website <http://earthquake.usgs.gov/hazards/> using the specific zip code or latitude and longitude that identify the location.

In regions other than those shown in the maps in **IBC-2012**, **FEMA 450-2**, or **ANSI/AWWA D115-17**, S_s and S_1 may be replaced by the maximum considered earthquake spectral response accelerations from 5 percent damped response spectra representing earthquakes with a 2 percent probability of exceedance in a 50-year period, equivalent to a recurrence interval of approximately 2500 years.